

Assignment 3

Your Name

2026-09-30

Goal of this assignment

The goals of this assignment are to:

- Review treatment structure and design structure of designed experiments.
- Analyze data generated by a designed experiment using statistical software.

Please complete exercises (1) and (2), rename the Rmd file to “Assignment2_YourLastName.Rmd” (e.g., “Assignment2_Smith.Rmd”), your name in the header, and knit the Rmd to an html file or pdf file. Please submit that html or pdf file on CANVAS by Wednesday, September 10th by midnight. You may work in pairs, but each one of you will have to submit your own file.

Formatting: Please make sure that your knitted pdf or html document does not include unnecessary warnings/messages.

For the statistical models, you can use L^AT_EXcode embedded in your Rmd, or paste a picture. You can check the raw Rmd files from the lectures (including L^AT_EXcode) at https://github.com/stat870/fall2026/tree/main/_notes. Otherwise, you can check out the Rmd cheatsheet or the `knitr::include_graphics()` function.

1.

Consider a simple randomized complete block design (RCBD) with 3 treatments and 2 blocks (total $n = 6$ observations).

1.a. If blocks are modeled as fixed effects, what do we expect for the standard error of the mean, and what do we expect for the standard error of mean differences?

1.b. If blocks are modeled as random effects ($u_j \sim N(0, \sigma_u^2)$), what do we expect for the standard error of the mean, and what do we expect for the standard error of mean differences?

1.c. If blocks are modeled as random effects, write out the theoretical covariance between observation 1 (Block 1, Trt A) and observation 2 (Block 1, Trt B). How does this differ from observation 1 (Block 1, Trt A) and observation 4 (Block 2, Trt A)?

1.d. You are discussing this problem with your colleague and they tell you that the blocks should be modeled as random, because they don’t care about the block effects. What would you tell them?

2.

The data below were generated by an aquaculture clinical field trial in the Bay of Fundy, Canada. The objective of this research was to compare the performance of different vaccines on the growth and survival of Atlantic salmon under standard production conditions. On day 0 of the study, 4 different vaccines were applied to a total of 100 randomly selected fish each. Salmon were individually tagged in February 2005 and followed through to harvest in August 2007. All fish were always kept in the same cage. Researchers recorded (1) presence of jaw deformity, and (2) weight throughout the trial. As jaw deformities are formed at an early stage of growth and are not healed, detection of a jaw deformity at any sampling event implied that the fish was labelled as having a jaw deformity throughout the entire growth period.

The outcome of interest is the weight, as weights increase from means of 60 g at day 0, to 5,700 g at harvest. However, it is also important to know whether there are differences in propensity to presenting jaw deformity among vaccines.

2.a. Describing the experiment design in terms of the treatment structure and design structure.

2.b. Analyze the data and determine: - If there are differences between vaccines, especially at harvest, - If there is a “better vaccine” for final weight, - If there is difference in jaw deformity between vaccine treatments.

```
library(tidyverse)
library(ggpubr)

url <- "https://raw.githubusercontent.com/stat870/fall2026/refs/heads/main/data/fish_vaccines.csv"

fish <- read.csv(url)

# plot the data
fish %>%
  ggplot(aes(day, wt))+
  geom_point(aes(fill = factor(vaccine),
                             group = factor(vaccine)),
             position = position_dodge(width = 60),
             shape=21, size =3.5, alpha =.7)+
  labs(y = expression(Weight~(g~fish^{-1})), x = "Time (days)")+
  theme_classic()+
  labs(fill="Vaccine treatment")+
  theme(legend.position = "bottom")+
  scico::scale_fill_scico_d()
```

